

Aegis Home System — Draft v5 V2.0

Adding Structural and Fastener Systems

Version 2.0 — June 2026

Iteration 5 hardens the structural skeleton — rebar, fasteners, and load-bearing connections.

Δ Change Log

- 3Y-TZP ZrO₂ fasteners: >1300 MPa tensile (2025 confirmed)
- IBC 2025 Section 2304 non-metallic fastener load-path documented
- BFRP rebar: ACI 440.11-25 seismic Type S provisions

New V5 Components

12. **Zirconia Structural Fasteners (CSMFAB0000000000010)** — 3Y-TZP ZrO₂ bolts; tensile >1300 MPa; IBC 2025 load-path compliant; no thermal bridge; GIC-invisible
13. **BFRP Rebar extended to full structure (CSMFAB0000000000009)** — Type S seismic application per ACI 440.11-25; 1000 MPa tensile; basalt/Elium® system recoverable via thermal solvolysis
14. **MAX Phase Ti₃AlC₂ inserts** for high-stress connection nodes — machinable, KNE-ductile, non-magnetic

V5 structural integrity: - Zero ferromagnetic elements in structure - No GIC eddy current heating in rebar under G5 storm - Explosive spalling risk: eliminated

V5 cumulative GIC threat reduction: ~89%